

Dimensional Bottleneck Compression, Variational Inference, and Latent Space Topology

A Unified Mathematical, Information-Theoretic, and Physical Foundation of
Autoencoders and Variational Autoencoders (VAE)

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1 The Autoencoder Paradigm and Deterministic Bottlenecks

Supervised learning maps input features $\mathbf{x} \in \mathbb{R}^{d_x}$ to targets $\mathbf{y} \in \mathbb{R}^{d_y}$. Conversely, **Unsupervised Representation Learning** seeks low-dimensional latent structures $\mathbf{z} \in \mathbb{R}^{d_z}$ ($d_z \ll d_x$) that capture the intrinsic manifold geometry of an unlabelled data distribution $p_{\text{data}}(\mathbf{x})$.

An **Autoencoder (AE)** is a neural architecture composed of two non-linear mappings: an **Encoder** f_ϕ that compresses high-dimensional inputs into a bottleneck vector, and a **Decoder** g_θ that reconstructs original input vectors from this latent coordinate.

Definition 1.1 (Deterministic Autoencoder Architecture). Let $\mathbf{x} \in \mathbb{R}^{d_x}$ be an input vector sampled from dataset $\mathcal{D}$. An Autoencoder defines the composite function $\hat{\mathbf{x}} = g_\theta(f_\phi(\mathbf{x}))$:

$$\text{Encoder Mapping: } \mathbf{z} = f_\phi(\mathbf{x}) = \sigma_e(\mathbf{W}_e \mathbf{x} + \mathbf{b}_e) \in \mathbb{R}^{d_z} \quad (1)$$

$$\text{Decoder Mapping: } \hat{\mathbf{x}} = g_\theta(\mathbf{z}) = \sigma_d(\mathbf{W}_d \mathbf{z} + \mathbf{b}_d) \in \mathbb{R}^{d_x} \quad (2)$$

where $\phi = \{\mathbf{W}_e, \mathbf{b}_e\}$ and $\theta = \{\mathbf{W}_d, \mathbf{b}_d\}$ are encoder and decoder parameters, σ_e, σ_d are non-linear activation functions, and $d_z \ll d_x$ establishes an informational **bottleneck**.

Parameters (ϕ^*, θ^*) are learned by minimizing an empirical reconstruction loss $\mathcal{L}_{\text{recon}}(\mathbf{x}, \hat{\mathbf{x}})$:

$$(\phi^*, \theta^*) = \arg \min_{\phi, \theta} \frac{1}{N} \sum_{i=1}^N \mathcal{L}_{\text{recon}}(\mathbf{x}^{(i)}, g_\theta(f_\phi(\mathbf{x}^{(i)}))) \quad (3)$$

Common reconstruction metrics include:

- **Mean Squared Error (MSE):** For continuous features $\mathbf{x} \in \mathbb{R}^{d_x}$:

$$\mathcal{L}_{\text{MSE}}(\mathbf{x}, \hat{\mathbf{x}}) = \frac{1}{2} \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 = \frac{1}{2} \sum_{j=1}^{d_x} (x_j - \hat{x}_j)^2 \quad (4)$$

- **Binary Cross-Entropy (BCE):** For normalized or binary features $\mathbf{x} \in [0, 1]^{d_x}$:

$$\mathcal{L}_{\text{BCE}}(\mathbf{x}, \hat{\mathbf{x}}) = - \sum_{j=1}^{d_x} [x_j \ln \hat{x}_j + (1 - x_j) \ln(1 - \hat{x}_j)] \quad (5)$$

1.1 Linear Autoencoders vs. Principal Component Analysis (PCA)

When activations are identity functions ($\sigma_e(u) = u, \sigma_d(u) = u$) and loss is MSE, autoencoders are closely related to linear dimensionality reduction methods.

Theorem 1.1 (Linear Autoencoder Equivalence to PCA (Baldi & Hornik, 1989)). Let $f_\phi(\mathbf{x}) = \mathbf{W}_e \mathbf{x}$ and $g_\theta(\mathbf{z}) = \mathbf{W}_d \mathbf{z}$ be linear maps with $\mathbf{W}_e \in \mathbb{R}^{d_z \times d_x}$ and $\mathbf{W}_d \in \mathbb{R}^{d_x \times d_z}$. For mean-centered data $X \in \mathbb{R}^{N \times d_x}$, global minima of the linear objective:

$$J(\mathbf{W}_e, \mathbf{W}_d) = \frac{1}{2N} \|X - X \mathbf{W}_e^T \mathbf{W}_d^T\|_F^2 \quad (6)$$

span the exact same subspace as the top d_z principal components of the empirical sample covariance matrix $\Sigma = \frac{1}{N} X^T X$.

While linear autoencoders span the PCA subspace, deep non-linear autoencoders (f_ϕ, g_θ as multi-layer networks) construct non-linear curved manifolds, enabling far higher compression ratios.

2 Variational Autoencoders (VAEs) and Generative Probabilistic Latent Spaces

Despite their compression utility, deterministic autoencoders exhibit severe pathologies when used as generative models: . ****Discontinuous Latent Geometry:**** Deterministic training places codes $\mathbf{z}^{(i)}$ at isolated points in $\mathbb{R}^{d_z}$ surrounded by unconstrained empty voids. Sampling arbitrary coordinates $\mathbf{z} \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ produces non-sensical reconstructions.

****Lack of Generative Density Control:**** The encoder does not enforce a structured prior distribution $p(\mathbf{z})$ over latent space.

****Variational Autoencoders (VAEs)****, introduced by Kingma & Welling (2013) and Rezende et al. (2014), solve this by modeling data generation through a ****Directed Probabilistic Latent Variable Model****.

2.1 Generative Model Formulation

Consider a directed probabilistic process where latent vector $\mathbf{z} \in \mathbb{R}^{d_z}$ is drawn from a simple prior $p(\mathbf{z})$, and data sample $\mathbf{x} \in \mathbb{R}^{d_x}$ is generated from a complex conditional distribution $p_{\theta}(\mathbf{x} \mid \mathbf{z})$ parameterized by decoder network θ :

$$\text{Latent Prior Distribution: } \mathbf{z} \sim p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_z}) \quad (7)$$

$$\text{Generative Likelihood: } \mathbf{x} \sim p_{\theta}(\mathbf{x} \mid \mathbf{z}) = \mathcal{N}(\boldsymbol{\mu}_{\theta}(\mathbf{z}), \text{diag}(\boldsymbol{\sigma}_{\theta}^2(\mathbf{z}))) \quad (8)$$

The true marginal likelihood (evidence) $p_{\theta}(\mathbf{x})$ requires integrating over all latent space paths:

$$p_{\theta}(\mathbf{x}) = \int_{\mathbb{R}^{d_z}} p_{\theta}(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z}) d\mathbf{z} \quad (9)$$

This marginal integral is mathematically ****intractable**** because evaluating $p_{\theta}(\mathbf{x} \mid \mathbf{z})$ across high-dimensional space requires computing infinite network passes. Consequently, the true posterior distribution $p_{\theta}(\mathbf{z} \mid \mathbf{x}) = \frac{p_{\theta}(\mathbf{x} \mid \mathbf{z}) p(\mathbf{z})}{p_{\theta}(\mathbf{x})}$ cannot be computed analytically.

To bypass this intractability, VAEs introduce a ****Variational Inference Network (Inference Encoder)**** $q_{\phi}(\mathbf{z} \mid \mathbf{x})$ parameterized by encoder network ϕ , designed to approximate the true posterior $p_{\theta}(\mathbf{z} \mid \mathbf{x})$.

3 Derivation of the Evidence Lower Bound (ELBO)

We derive the objective function of Variational Autoencoders—the ****Evidence Lower Bound (ELBO)****—by decomposing the marginal log-likelihood $\ln p_{\theta}(\mathbf{x})$.

Theorem 3.1 (ELBO Mathematical Decomposition). For any arbitrary density $q_{\phi}(\mathbf{z} \mid \mathbf{x})$, the marginal log-likelihood decomposes into:

$$\ln p_{\theta}(\mathbf{x}) = \text{ELBO}(\theta, \phi; \mathbf{x}) + D_{\text{KL}}(q_{\phi}(\mathbf{z} \mid \mathbf{x}) \parallel p_{\theta}(\mathbf{z} \mid \mathbf{x})) \quad (10)$$

where the Evidence Lower Bound $\text{ELBO}(\theta, \phi; \mathbf{x})$ is defined as:

$$\text{ELBO}(\theta, \phi; \mathbf{x}) \equiv \mathbb{E}_{q_{\phi}(\mathbf{z} \mid \mathbf{x})} [\ln p_{\theta}(\mathbf{x} \mid \mathbf{z})] - D_{\text{KL}}(q_{\phi}(\mathbf{z} \mid \mathbf{x}) \parallel p(\mathbf{z})) \quad (11)$$

Proof. Expand the log marginal likelihood using expectations over variational posterior $q_\phi(\mathbf{z} | \mathbf{x})$:

$$\begin{aligned}
\ln p_\theta(\mathbf{x}) &= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\ln p_\theta(\mathbf{x})] = \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln p_\theta(\mathbf{x}) d\mathbf{z} \\
&= \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{p_\theta(\mathbf{z} | \mathbf{x})} \right) d\mathbf{z} \\
&= \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z} | \mathbf{x})} \cdot \frac{q_\phi(\mathbf{z} | \mathbf{x})}{p_\theta(\mathbf{z} | \mathbf{x})} \right) d\mathbf{z} \\
&= \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{p_\theta(\mathbf{x}, \mathbf{z})}{q_\phi(\mathbf{z} | \mathbf{x})} \right) d\mathbf{z} + \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{q_\phi(\mathbf{z} | \mathbf{x})}{p_\theta(\mathbf{z} | \mathbf{x})} \right) d\mathbf{z} \\
&= \mathcal{F}(\theta, \phi; \mathbf{x}) + D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{x}) \| p_\theta(\mathbf{z} | \mathbf{x}))
\end{aligned} \tag{12}$$

Expanding term $\mathcal{F}(\theta, \phi; \mathbf{x})$ using Bayes' relation $p_\theta(\mathbf{x}, \mathbf{z}) = p_\theta(\mathbf{x} | \mathbf{z})p(\mathbf{z})$:

$$\begin{aligned}
\mathcal{F}(\theta, \phi; \mathbf{x}) &= \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{p_\theta(\mathbf{x} | \mathbf{z})p(\mathbf{z})}{q_\phi(\mathbf{z} | \mathbf{x})} \right) d\mathbf{z} \\
&= \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln p_\theta(\mathbf{x} | \mathbf{z}) d\mathbf{z} + \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{p(\mathbf{z})}{q_\phi(\mathbf{z} | \mathbf{x})} \right) d\mathbf{z} \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\ln p_\theta(\mathbf{x} | \mathbf{z})] - \int_{\mathbb{R}^{d_z}} q_\phi(\mathbf{z} | \mathbf{x}) \ln \left(\frac{q_\phi(\mathbf{z} | \mathbf{x})}{p(\mathbf{z})} \right) d\mathbf{z} \\
&= \mathbb{E}_{q_\phi(\mathbf{z}|\mathbf{x})} [\ln p_\theta(\mathbf{x} | \mathbf{z})] - D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z}))
\end{aligned} \tag{13}$$

This completes the exact derivation of the ELBO.

Because Kullback-Leibler divergence is non-negative ($D_{\text{KL}} \geq 0$), the ELBO forms a strict mathematical lower bound on data log-evidence:

$$\ln p_\theta(\mathbf{x}) \geq \text{ELBO}(\theta, \phi; \mathbf{x}) \tag{14}$$

Maximizing the ELBO achieves two simultaneous objectives: 1. It maximizes data log-likelihood $\ln p_\theta(\mathbf{x})$ (improving reconstruction quality). 2. It minimizes $D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{x}) \| p_\theta(\mathbf{z} | \mathbf{x}))$, forcing variational posterior q_ϕ to converge to true posterior p_θ .

3.1 Analytical Closed-Form KL Divergence for Gaussian Distributions

In standard VAEs, the variational posterior is modeled as a diagonal Gaussian distribution:

$$q_\phi(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\mathbf{z} | \boldsymbol{\mu}_\phi(\mathbf{x}), \text{diag}(\boldsymbol{\sigma}_\phi^2(\mathbf{x}))) \tag{15}$$

and the latent prior is a standard multivariate Gaussian $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_z})$.

Theorem 3.2 (Closed-Form Gaussian KL Divergence). For d_z -dimensional diagonal Gaussian $q_\phi(\mathbf{z} | \mathbf{x}) = \mathcal{N}(\boldsymbol{\mu}, \text{diag}(\boldsymbol{\sigma}^2))$ and isotropic prior $p(\mathbf{z}) = \mathcal{N}(\mathbf{0}, \mathbf{I})$, the KL divergence evaluates analytically as:

$$D_{\text{KL}}(q_\phi(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z})) = -\frac{1}{2} \sum_{j=1}^{d_z} (1 + \ln(\sigma_j^2) - \mu_j^2 - \sigma_j^2) \tag{16}$$

Proof. Expand the definition of KL divergence between two multivariate continuous distributions:

$$D_{\text{KL}}(q \| p) = \int q(\mathbf{z}) \ln q(\mathbf{z}) d\mathbf{z} - \int q(\mathbf{z}) \ln p(\mathbf{z}) d\mathbf{z} = \mathbb{E}_q[\ln q(\mathbf{z})] - \mathbb{E}_q[\ln p(\mathbf{z})] \tag{17}$$

Evaluating expected log-densities:

$$\begin{aligned}
\mathbb{E}_q[\ln q(\mathbf{z})] &= -\frac{d_z}{2} \ln(2\pi) - \frac{1}{2} \sum_{j=1}^{d_z} \ln(\sigma_j^2) - \frac{1}{2} \sum_{j=1}^{d_z} \mathbb{E}_q \left[\frac{(z_j - \mu_j)^2}{\sigma_j^2} \right] \\
&= -\frac{d_z}{2} \ln(2\pi) - \frac{1}{2} \sum_{j=1}^{d_z} \ln(\sigma_j^2) - \frac{d_z}{2}
\end{aligned} \tag{18}$$

$$\begin{aligned}
\mathbb{E}_q[\ln p(\mathbf{z})] &= -\frac{d_z}{2} \ln(2\pi) - \frac{1}{2} \sum_{j=1}^{d_z} \mathbb{E}_q[z_j^2] \\
&= -\frac{d_z}{2} \ln(2\pi) - \frac{1}{2} \sum_{j=1}^{d_z} (\mu_j^2 + \sigma_j^2)
\end{aligned} \tag{19}$$

Subtracting $\mathbb{E}_q[\ln p(\mathbf{z})]$ from $\mathbb{E}_q[\ln q(\mathbf{z})]$ yields the analytical result:

$$D_{\text{KL}} = \frac{1}{2} \sum_{j=1}^{d_z} (\mu_j^2 + \sigma_j^2 - 1 - \ln(\sigma_j^2)) = -\frac{1}{2} \sum_{j=1}^{d_z} (1 + \ln(\sigma_j^2) - \mu_j^2 - \sigma_j^2) \tag{20}$$

□

4 The Reparameterization Trick

To optimize the ELBO using backpropagation, we must evaluate parameter gradients of the expected reconstruction loss:

$$\nabla_{\phi} \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\ln p_{\theta}(\mathbf{x} | \mathbf{z})] \tag{21}$$

If $\mathbf{z}$ is sampled directly from the stochastic distribution $\mathbf{z} \sim q_{\phi}(\mathbf{z} | \mathbf{x})$, parameter vector ϕ governs the distribution itself. Stochastic sampling acts as a non-differentiable operation, blocking backpropagation gradient flow through the sampler node ($\frac{\partial \mathbf{z}}{\partial \phi}$ is undefined).

4.1 Isolating Stochasticity via Auxiliary Noise

Kingma & Welling (2013) introduced the **Reparameterization Trick**, expressing random variable $\mathbf{z}$ as a deterministic, differentiable function of input $\mathbf{x}$, encoder parameters $(\mu_{\phi}, \sigma_{\phi})$, and an independent auxiliary noise vector ϵ :

Definition 4.1 (Gaussian Reparameterization Mapping). Let $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}_{d_z})$ be an independent parameter-free noise vector. The stochastic latent vector $\mathbf{z}$ is reparameterized deterministically as:

$$\mathbf{z} = g_{\phi}(\mathbf{x}, \epsilon) \equiv \mu_{\phi}(\mathbf{x}) + \sigma_{\phi}(\mathbf{x}) \odot \epsilon \tag{22}$$

where $\sigma_{\phi}(\mathbf{x}) = \exp\left(\frac{1}{2} \ln \sigma_{\phi}^2(\mathbf{x})\right)$ for numerical stability.

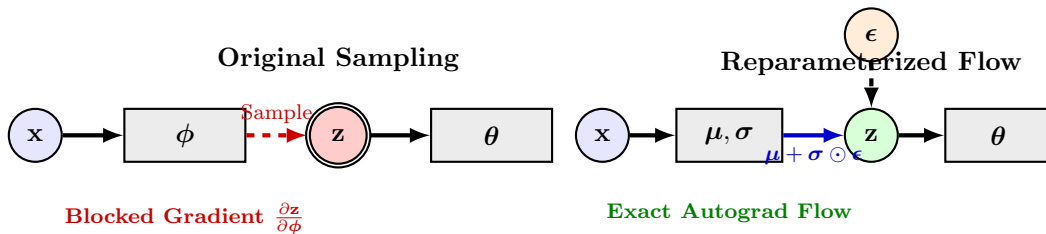


Figure 1: Original non-differentiable stochastic sampling vs. the Reparameterization Trick, which constructs an unobstructed computational path for backpropagation.

Applying reparameterization converts expectation gradients into standard Monte Carlo path gradients:

$$\nabla_{\phi} \mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [\ln p_{\theta}(\mathbf{x} | \mathbf{z})] = \mathbb{E}_{p(\epsilon)} [\nabla_{\phi} \ln p_{\theta}(\mathbf{x} | g_{\phi}(\mathbf{x}, \epsilon))] \tag{23}$$

Using a single Monte Carlo sample $\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ per minibatch sample yields an unbiased, low-variance estimator of the gradient that can be evaluated directly via backpropagation.

5 Information-Theoretic and Physical Analogies

5.1 Information-Theoretic View: Rate-Distortion Theory and Minimum Description Length

In Rate-Distortion Theory (Shannon, 1948), the VAE objective optimizes a lossy data compression framework:

$$\mathcal{L}_{\text{VAE}} = \underbrace{\mathbb{E}_{q_{\phi}(\mathbf{z}|\mathbf{x})} [-\ln p_{\theta}(\mathbf{x} | \mathbf{z})]}_{\text{Distortion } D} + \beta \underbrace{D_{\text{KL}}(q_{\phi}(\mathbf{z} | \mathbf{x}) \| p(\mathbf{z}))}_{\text{Rate } R} \quad (24)$$

* **Distortion D (Reconstruction Error):** Measures expected code reconstruction infidelity under the noise channel. * **Rate R (KL Divergence):** Measures the information rate (in nats or bits) required to transmit code $\mathbf{z}$ relative to standard Gaussian prior baseline $p(\mathbf{z})$. * **β -VAE Trade-off:** Scaling hyperparameter $\beta > 1$ (Higgins et al., 2017) forces the latent space to prioritize rate compression over reconstruction, driving structural **disentanglement** along orthogonal factor axes.

5.2 Physical Analogy: Helmholtz Free Energy Minimization

The VAE ELBO matches the formulation of **Helmholtz Free Energy** in statistical thermodynamics.

Consider a multi-particle thermodynamic ensemble with state configurations $\mathbf{z}$ and observed macroscopic properties $\mathbf{x}$. The Helmholtz Free Energy F at thermal equilibrium is:

$$F = U - TS \quad (25)$$

where U is internal system energy, S is thermodynamic entropy, and T is absolute temperature.

Mapping thermodynamic variables directly to the negative ELBO:

- **Internal Energy $U \leftrightarrow -\mathbb{E}_q[\ln p_{\theta}(\mathbf{x} | \mathbf{z})]$:** Minimizing U configuration energy corresponds to minimizing structural field reconstruction errors.
- **Entropic Penalty $TS \leftrightarrow D_{\text{KL}}(q_{\phi} \| p)$:** The KL divergence acts as an entropic constraint preventing thermal collapse into zero-variance microstates, enforcing smooth density distributions across the latent space manifold.

Maximizing the ELBO corresponds to finding a thermodynamic equilibrium state that minimizes total Helmholtz Free Energy.

6 Production-Grade PyTorch Implementation

The following script provides a production-grade PyTorch implementation of a Variational Autoencoder (VAE). It includes a bottleneck architecture, the exact reparameterization trick, Gaussian KL-divergence, composite ELBO loss evaluation, a training pipeline, and latent space generation.

```
1 import torch
2 import torch.nn as nn
3 import torch.nn.functional as F
4 import numpy as np
5
6 # Set reproducibility seeds
7 torch.manual_seed(42)
8 np.random.seed(42)
9
10 class VariationalAutoencoder(nn.Module):
11     """
12     Production-grade Variational Autoencoder (VAE) supporting Gaussian
```

```

13     Reparameterization, Analytical KL Divergence, and Generative Latent
14     Sampling.
15     """
16     def __init__(self, input_dim: int = 784, hidden_dim: int = 256,
17     latent_dim: int = 32):
18         super(VariationalAutoencoder, self).__init__()
19         self.input_dim = input_dim
20         self.latent_dim = latent_dim
21
22         # --- ENCODER BACKBONE ---
23         self.encoder_fc = nn.Linear(input_dim, hidden_dim)
24         self.fc_mu = nn.Linear(hidden_dim, latent_dim)
25         self.fc_logvar = nn.Linear(hidden_dim, latent_dim)
26
27         # --- DECODER BACKBONE ---
28         self.decoder_fc = nn.Linear(latent_dim, hidden_dim)
29         self.decoder_out = nn.Linear(hidden_dim, input_dim)
30
31     def encode(self, x: torch.Tensor):
32         """Maps input x to variational Gaussian parameters (mu, log_var)."""
33         h = F.relu(self.encoder_fc(x))
34         mu = self.fc_mu(h)
35         log_var = self.fc_logvar(h) # Log-variance for numerical stability
36         return mu, log_var
37
38     def reparameterize(self, mu: torch.Tensor, log_var: torch.Tensor) ->
39     torch.Tensor:
40         """
41         Differentiable Reparameterization Trick:
42         z = mu + sigma * epsilon, where epsilon ~ N(0, I)
43         """
44         if self.training:
45             std = torch.exp(0.5 * log_var)
46             eps = torch.randn_like(std) # Independent noise sampler
47             return mu + eps * std
48         else:
49             return mu # Deterministic mean during evaluation
50
51     def decode(self, z: torch.Tensor) -> torch.Tensor:
52         """Maps latent coordinate z back to reconstructed data space x_hat.
53         """
54         h = F.relu(self.decoder_fc(z))
55         return torch.sigmoid(self.decoder_out(h)) # Sigmoid for normalized
56         [0, 1] data
57
58     def forward(self, x: torch.Tensor):
59         """Full forward pass: Encode -> Reparameterize -> Decode."""
60         mu, log_var = self.encode(x)
61         z = self.reparameterize(mu, log_var)
62         x_recon = self.decode(z)
63         return x_recon, mu, log_var
64
65     def vae_loss_function(x_recon: torch.Tensor, x: torch.Tensor,
66     mu: torch.Tensor, log_var: torch.Tensor, beta: float
67     = 1.0):
68         """
69         Evaluates Composite Loss = Reconstruction Loss + Beta * KL Divergence.
70         Negative ELBO Minimization.

```

```

66     """
67     # 1. Binary Cross-Entropy Reconstruction Loss (summed over feature
dimension)
68     bce_loss = F.binary_cross_entropy(x_recon, x, reduction='sum')
69
70     # 2. Analytical Closed-Form Gaussian KL Divergence:
71     # D_KL = -0.5 * sum(1 + log(sigma^2) - mu^2 - sigma^2)
72     kl_divergence = -0.5 * torch.sum(1.0 + log_var - mu.pow(2) - log_var.
exp())
73
74     # Total batch loss averaged over batch dimension
75     batch_size = x.size(0)
76     total_loss = (bce_loss + beta * kl_divergence) / batch_size
77     return total_loss, bce_loss / batch_size, kl_divergence / batch_size
78
79
80 if __name__ == "__main__":
81     # Hyperparameters
82     B_batch = 64
83     d_in = 784          # e.g., 28x28 flattened image
84     d_hidden = 256
85     d_latent = 20
86     epochs = 10
87     learning_rate = 1e-3
88
89     # Generate Synthetic Normalized Data Matrix [0, 1]
90     X_synthetic = torch.rand((1000, d_in))
91
92     # Initialize VAE Model and Adam Optimizer
93     vae = VariationalAutoencoder(input_dim=d_in, hidden_dim=d_hidden,
latent_dim=d_latent)
94     optimizer = torch.optim.Adam(vae.parameters(), lr=learning_rate)
95
96     print("=== VARIATIONAL AUTOENCODER (VAE) INITIALIZATION SUCCESSFUL ==="
)
97     print(f"Data Feature Dimension : {d_in}")
98     print(f"Latent Bottleneck Dim   : {d_latent}")
99     print(f"Model Parameter Count   : {sum(p.numel() for p in vae.parameters
())}\n")
100
101     # Training Loop
102     vae.train()
103     for epoch in range(1, epochs + 1):
104         num_batches = int(np.ceil(len(X_synthetic) / B_batch))
105         epoch_loss = 0.0
106
107         for b in range(num_batches):
108             x_batch = X_synthetic[b * B_batch : (b + 1) * B_batch]
109
110             optimizer.zero_grad()
111             x_recon, mu, log_var = vae(x_batch)
112
113             loss, bce, kl = vae_loss_function(x_recon, x_batch, mu, log_var
, beta=1.0)
114             loss.backward()
115             optimizer.step()
116
117             epoch_loss += loss.item() * len(x_batch)
118
119         avg_loss = epoch_loss / len(X_synthetic)

```



```

120     print(f"Epoch [{epoch:02d}/{epochs}] - Loss: {avg_loss:.4f} | Final
Batch BCE: {bce.item():.2f} | KL: {kl.item():.2f}")
121
122     # Demonstration of Generative Sampling from Prior N(0, I)
123     vae.eval()
124     with torch.no_grad():
125         z_prior_samples = torch.randn(5, d_latent) # Sample 5 random latent
vectors
126         generated_samples = vae.decode(z_prior_samples)
127
128     print(f"\nGenerative Sampling Demo:")
129     print(f"Prior Latent Samples Shape : {z_prior_samples.shape}")
130     print(f"Generated Data Shape       : {generated_samples.shape}")

```

Listing 1: Production-grade PyTorch implementation of a Variational Autoencoder (VAE).